What changed in skill forecasting—and how much?

Tactical observer B01 → frozen trained pilot A01 | Exact +1, +2 and +5 s selections

55.90 → 63.48%

+5 s TOP-1

+7.58 percentage points
17.19% fewer top-1 errors

0.274 → 0.366

+5 s MACRO-F1

Equal weight across all 33 labels
The gain extends beyond overall accuracy

THE COMPARISON

Training frequency → original GRU / transformer → improved 12-model ensemble

The prediction target and evaluation population stay fixed

Same 33-label vocabulary, same pilots, same scenarios and same missing-label masks.

ONE-SIDED INPUT

B01 observations available through time t.
No pilot ID or private selected label as input.
Labels are the exact choices at t+1, t+2, t+5.

GROUPED DATA

3,000 episodes: 300 scenarios × 10 pilots.
210 train / 45 validation / 45 test groups.
All ten pilot encounters stay in one group.

Forecast horizon	Valid validation origins	Labels with truth support
+1 s	30,802	26 of 33
+2 s	30,466	26 of 33
+5 s	29,458	26 of 33

The improved system has a larger information budget

The experiment compares complete forecasting procedures, not architecture alone.

Setting	Original GRU	Original transformer	Improved ensemble
Observation history	5 seconds	5 seconds	20 / 60 seconds
Features per token	54	54	91
Neural architecture	2-layer GRU	2-layer transformer	4 temporal MLP variants
Parameters	234,723	439,651	28,977,060 total
Seeds / fitted models	Seed 42 / one	Seed 42 / one	42, 43, 44 / twelve
Checkpoint selection	Validation NLL	Validation NLL	Training-only partitions
Selected epochs	15	20	46 / 32 / 40 / 40

Frequency reference: training-label priors only; no observation history or neural network.

Read accuracy, class balance and confidence together

Each metric answers a different question; a single ranking can hide important trade-offs.

Metric	Question it answers
Top-1 / top-3 ↑	Is the true label first / among the three highest probabilities?
Macro-F1 (33) ↑	How well are labels recovered when every catalogue entry counts equally?
Supported macro-F1 (26) ↑	The same F1 average, restricted to labels with truth support.
Negative log-likelihood ↓	How much probability is assigned to the actual label?
Multiclass Brier score ↓	How far is the entire probability vector from the correct one-hot target?
Calibration error (ECE) ↓	How close is top-label confidence to observed accuracy in fixed bins?

+1 second: all four methods on the same origins

30,802 valid origins | Higher accuracy / F1, lower NLL are better.

Model	Top-1 ↑	Top-3 ↑	F1 (33) ↑	F1 (26) ↑	NLL ↓
Training frequency	17.84%	50.03%	0.0092	0.0116	2.3921
Original GRU	54.40%	86.95%	0.2374	0.3013	1.2606
Original transformer	58.97%	88.40%	0.3010	0.3820	1.2216
Improved ensemble	65.62%	90.10%	0.4120	0.5230	1.1155

IMPROVEMENT OVER THE ORIGINAL TRANSFORMER

Top-1: +6.65 points · Top-3: +1.70 points

Macro-F1: +0.1110 · NLL: -0.1061

+2 seconds: all four methods on the same origins

30,466 valid origins | Higher accuracy / F1, lower NLL are better.

Model	Top-1 ↑	Top-3 ↑	F1 (33) ↑	F1 (26) ↑	NLL ↓
Training frequency	17.99%	50.41%	0.0092	0.0117	2.3899
Original GRU	54.01%	86.72%	0.2309	0.2931	1.2768
Original transformer	58.38%	88.22%	0.2953	0.3748	1.2361
Improved ensemble	65.17%	89.82%	0.4115	0.5223	1.1324

IMPROVEMENT OVER THE ORIGINAL TRANSFORMER

Top-1: +6.79 points · Top-3: +1.60 points

Macro-F1: +0.1162 · NLL: -0.1037

+5 seconds: all four methods on the same origins

29,458 valid origins | Higher accuracy / F1, lower NLL are better.

Model	Top-1 ↑	Top-3 ↑	F1 (33) ↑	F1 (26) ↑	NLL ↓
Training frequency	18.46%	51.48%	0.0094	0.0120	2.3799
Original GRU	51.56%	85.04%	0.2123	0.2694	1.3363
Original transformer	55.90%	86.51%	0.2739	0.3476	1.2903
Improved ensemble	63.48%	89.25%	0.3665	0.4651	1.1642

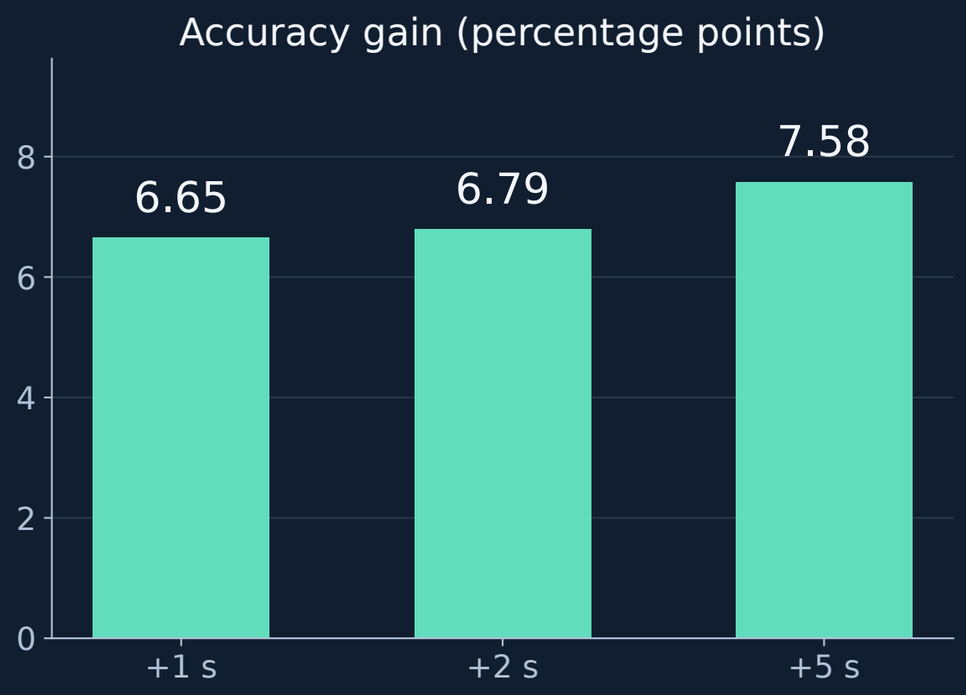
IMPROVEMENT OVER THE ORIGINAL TRANSFORMER

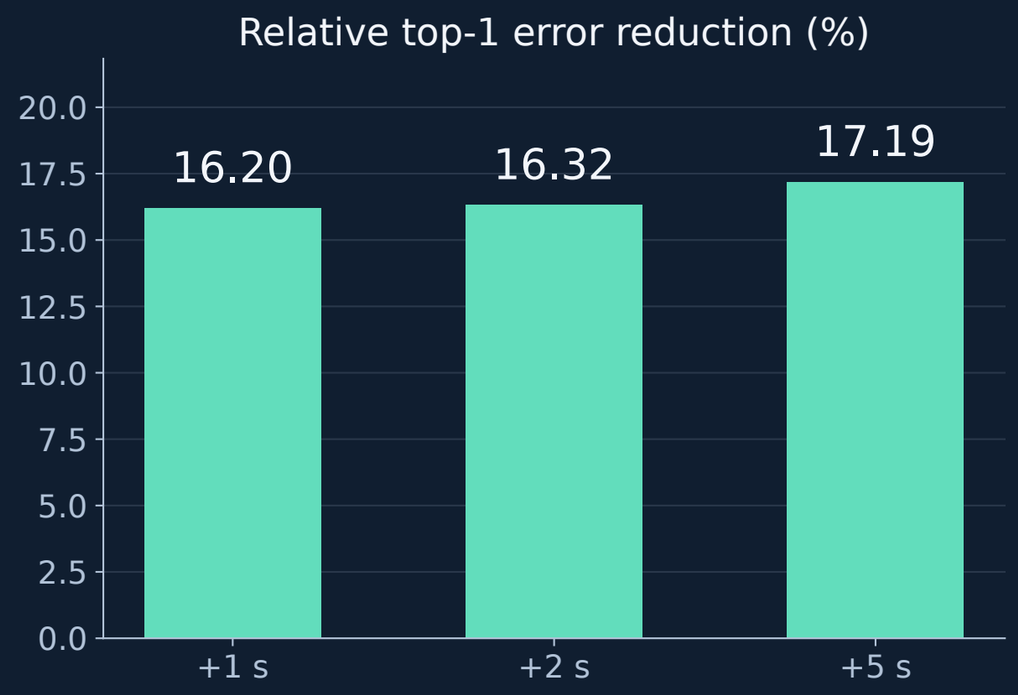
Top-1: +7.58 points · Top-3: +2.73 points

Macro-F1: +0.0926 · NLL: -0.1261

Percentage-point gain is not relative error reduction

Both improve, but their denominators differ.





+5 s: error rate falls from 44.10% to 36.52%; 7.58 ÷ 44.10 = 17.19% relative reduction.

Derived from matched top-1 accuracies | Development validation; test reserved

The ensemble fixes errors and introduces new ones

Paired outcomes on the same forecast origins reveal what the net accuracy gain hides.

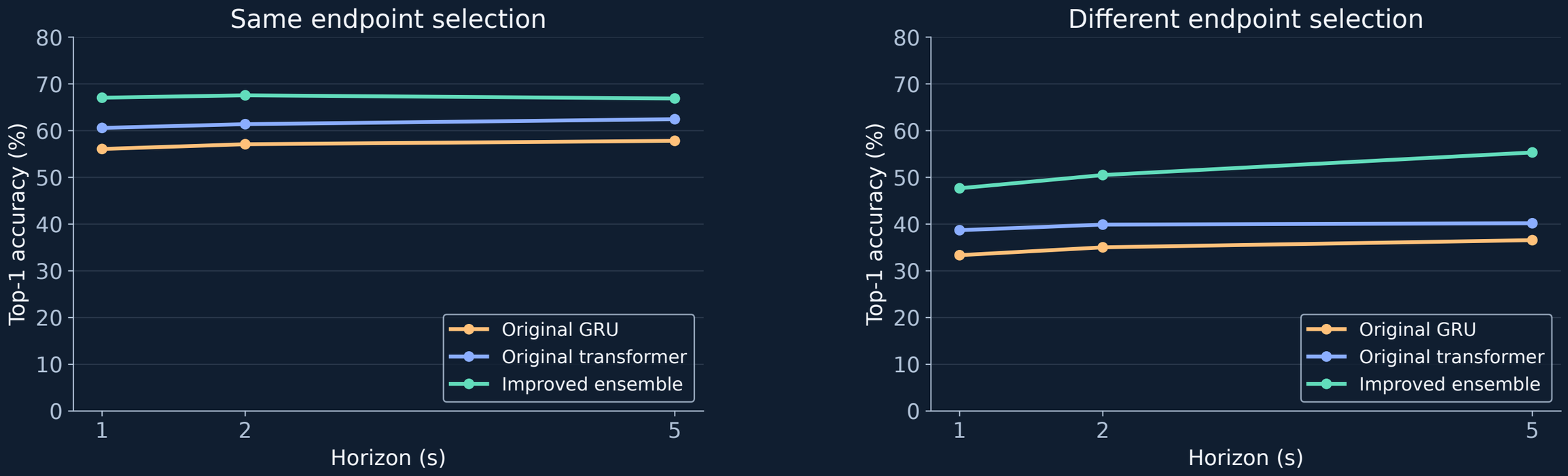
Horizon	Origins	Both correct	Error corrected	New error	Both wrong
+1 s	30,802	16,243	3,968	1,920	8,671
+2 s	30,466	15,872	3,983	1,914	8,697
+5 s	29,458	14,556	4,144	1,911	8,847

AT +5 SECONDS

4,144 corrections − 1,911 regressions = 2,233 additional correct forecasts.
8,847 origins are still wrong under both models.

Changed selections gain most; unchanged cases also matter

True current labels define these diagnostic subsets only; they are not inputs.

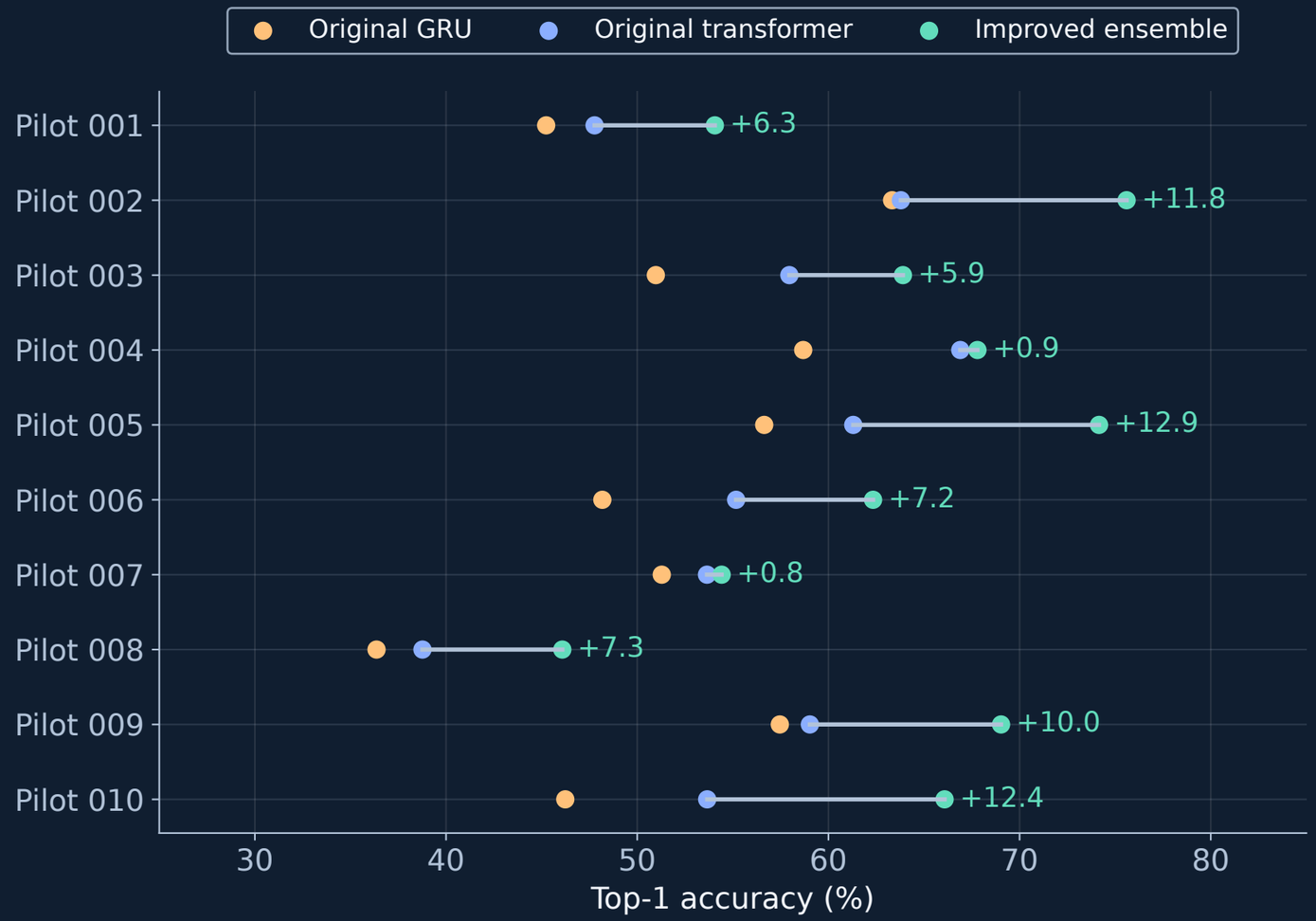


+5 s unchanged: 62.44% → 66.87% | changed: 40.18% → 55.33%

Unchanged +5 s top-3 falls 91.12% → 90.81%; its NLL worsens 1.0575 → 1.0813.

Every pilot improves; absolute difficulty still varies

+5-second top-1 accuracy for the known pilot population.



GAIN AT EVERY HORIZON

30 of 30 pilot / horizon cells improve.
At +5 s, gains range from
+0.77 to +12.86 points.

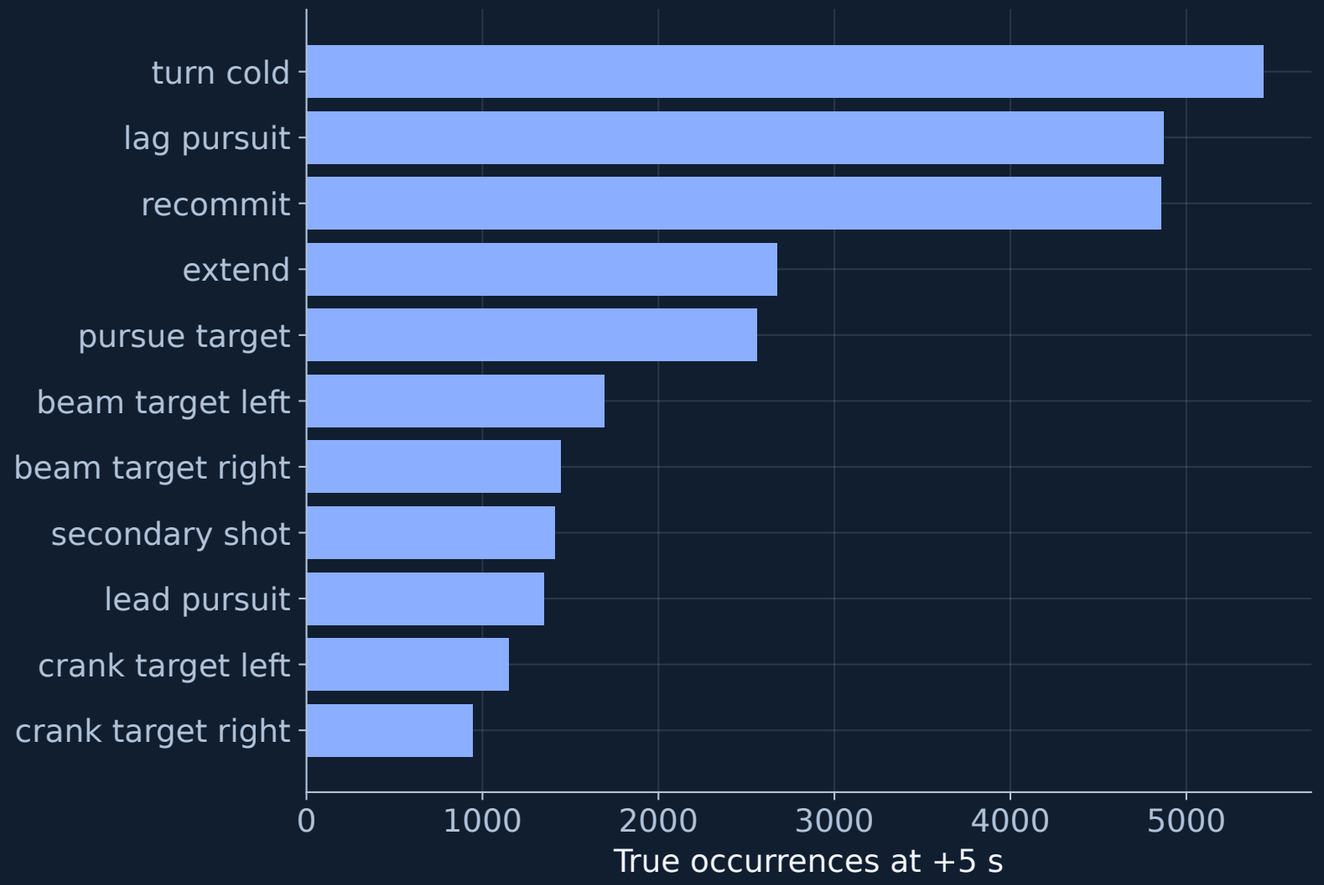
WHAT THIS DOES NOT MEASURE

Pilot competence or tactical quality.
Generalisation to unseen policies.
Significance for each pilot separately.

summary_metrics.csv | Per-pilot all cohort | Development validation; test reserved

A small set of labels dominates the validation windows

Class support explains why overall accuracy and macro-F1 tell different stories.



11 FREQUENT LABELS

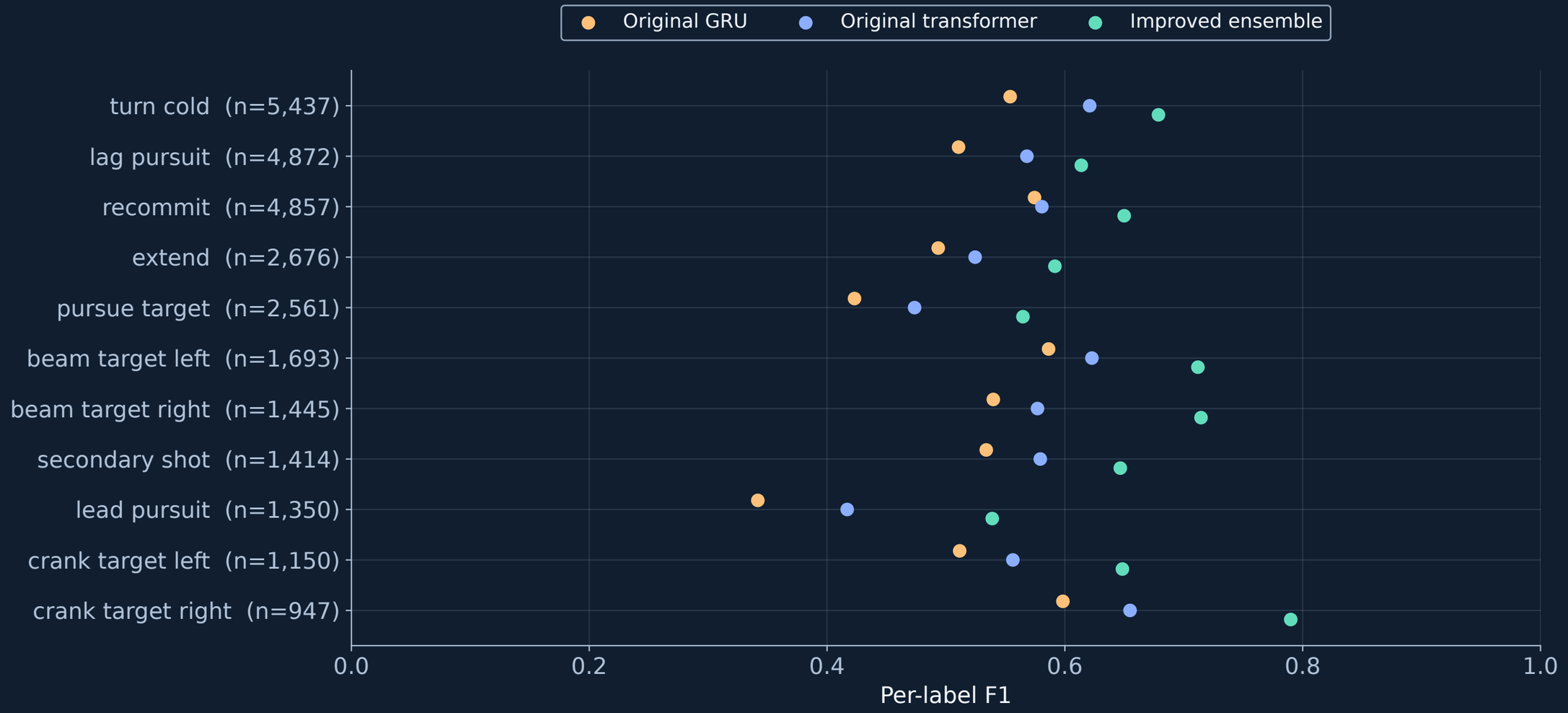
96.42% of valid windows.
At least 500 occurrences each.

THE REMAINDER

15 supported labels below 500.
7 entries with no truth support.
Rare-label estimates are fragile.

Frequent labels: compare F1 with the original models

All labels with at least 500 true +5 s occurrences; selected by support, not improvement.



Sparse labels: show the gains alongside the small counts

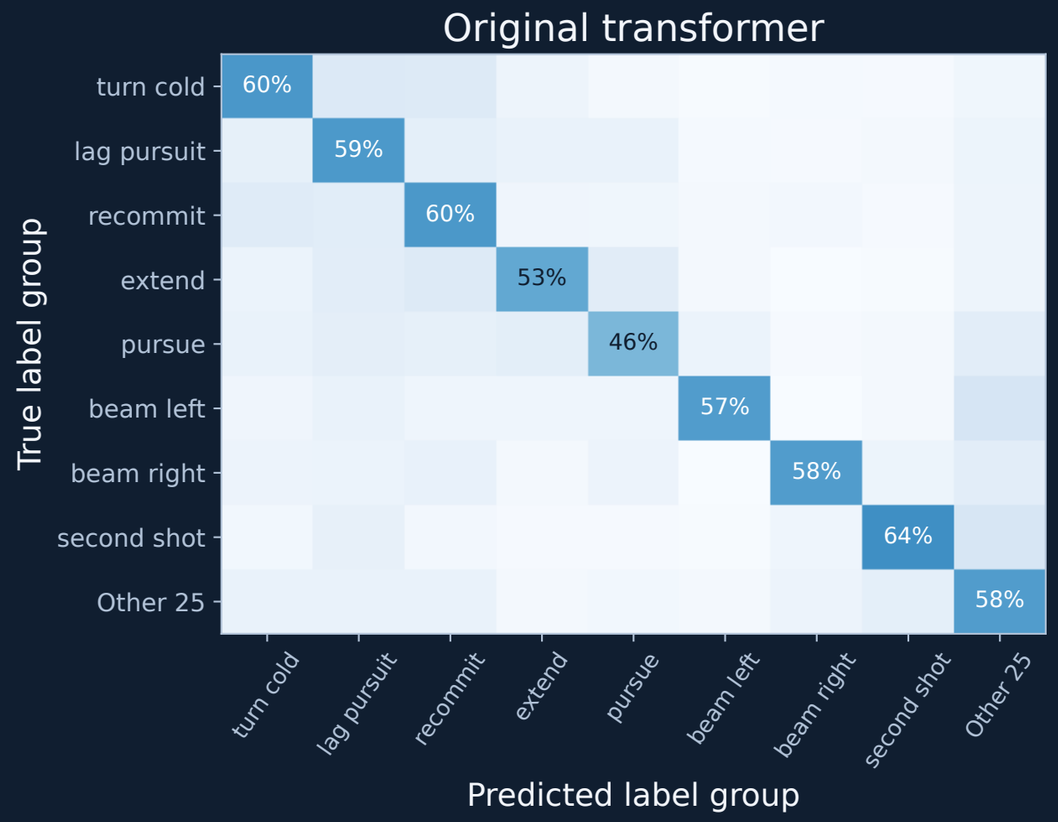
Every supported +5 s label below 500 occurrences. Do not infer reliable rare-class generalisation.

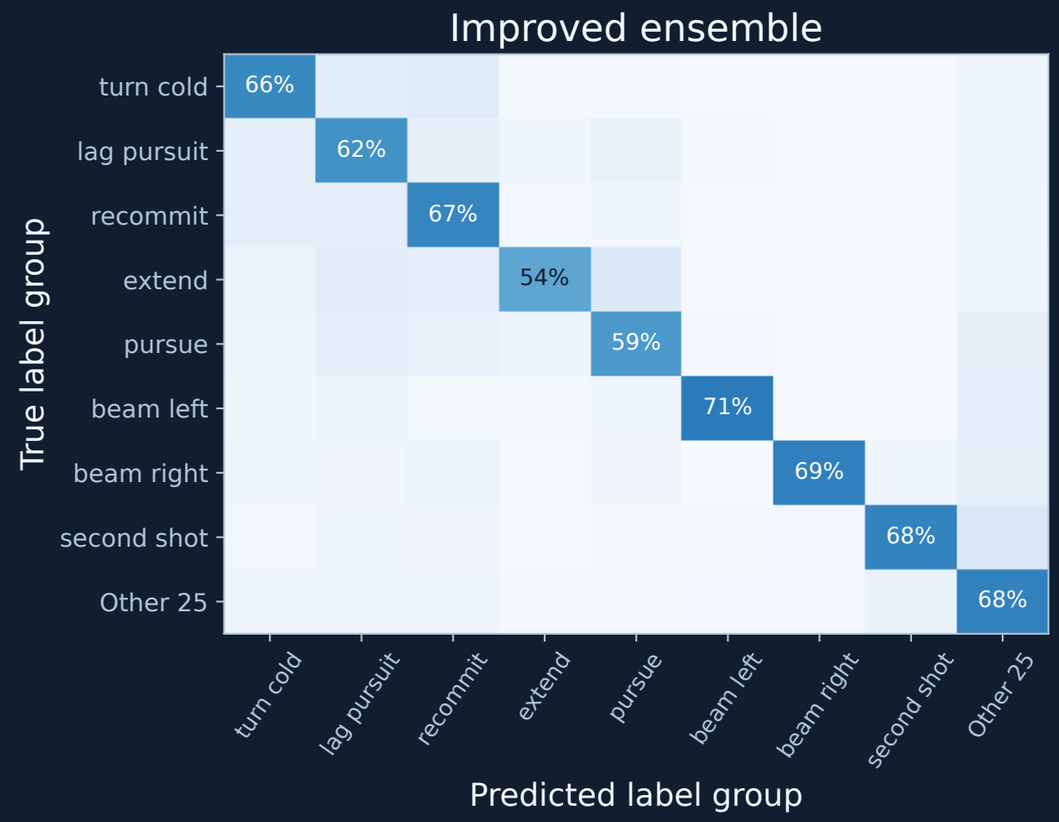
Label	Support	Old F1	New F1	New precision	New recall
short range attack	216	0.185	0.305	0.402	0.245
descend to altitude	162	0.480	0.619	0.528	0.747
launch	130	0.300	0.494	0.552	0.446
decelerate to speed	126	0.246	0.326	0.517	0.238
notch left	124	0.302	0.400	0.488	0.339
abort support	113	0.293	0.521	0.561	0.487
notch right	80	0.197	0.444	0.703	0.325
maintain heading	36	0.447	0.553	0.525	0.583
turn to heading	20	0.412	0.510	0.419	0.650
beam missile right	20	0.000	0.203	0.122	0.600
support missile	19	0.000	0.320	0.667	0.211
beam missile left	6	0.000	0.250	0.500	0.167
search	2	0.000	0.000	0.000	0.000
climb to altitude	1	0.000	0.000	0.000	0.000
defensive reversal	1	0.000	0.000	0.000	0.000

Three labels with only 1–2 true occurrences still have zero F1; seven further entries have zero support.

Confusion matrices reveal the remaining label mixtures

+5 s, row-normalized over all predictions. The eight most frequent true labels are shown individually.

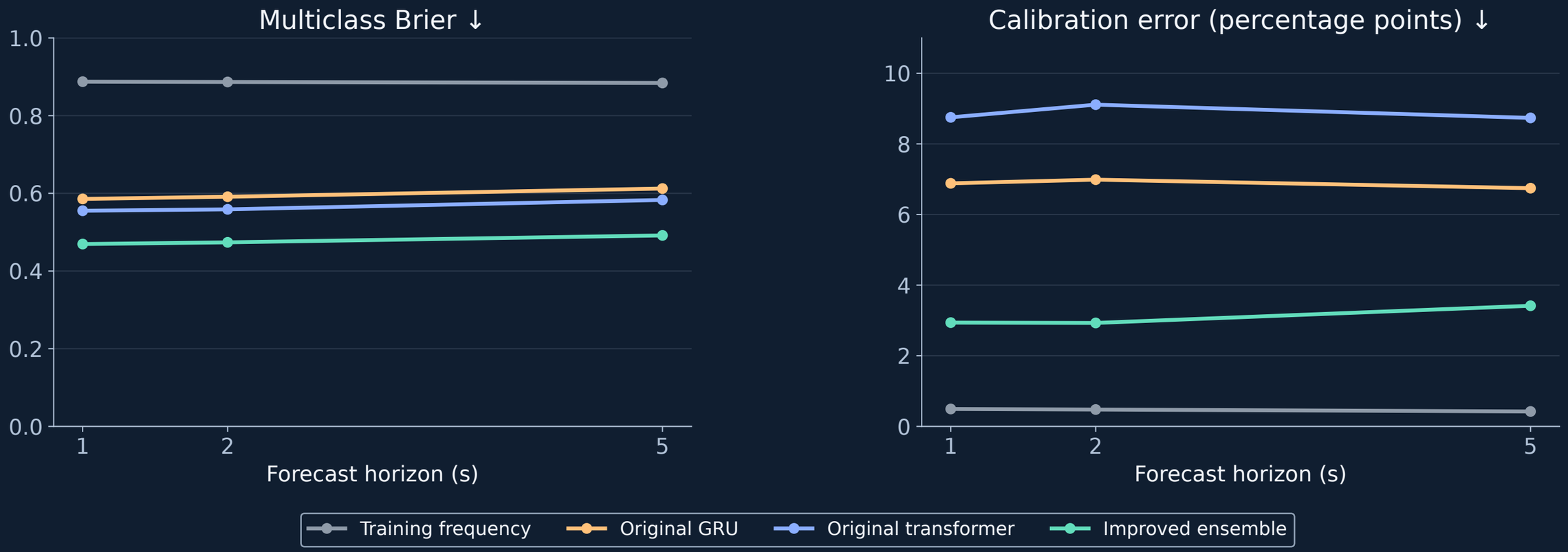




Darker = larger share of a true-label row. “Other 25” is for display only; scores retain all 33 labels.

The ensemble improves probability scores over both networks

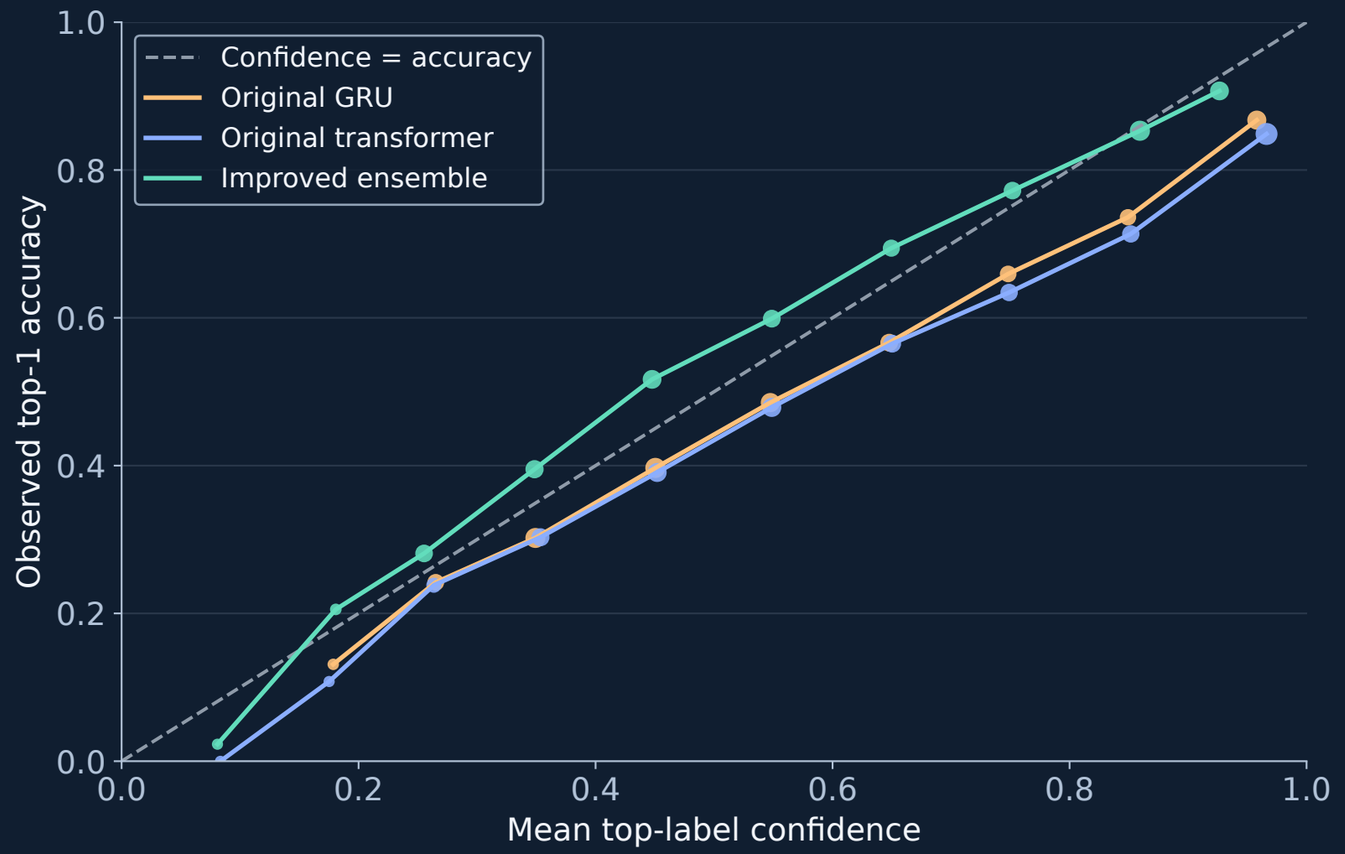
Brier measures the entire distribution; ECE measures binned top-label calibration.



Low ECE alone is not enough: the frequency reference has low ECE but poor accuracy.

Confidence is closer to observed accuracy after improvement

+5 s reliability diagram using ten fixed confidence bins; marker size reflects bin support.



AT +5 SECONDS

Original transformer
Mean confidence 64.64% / accuracy 55.90%

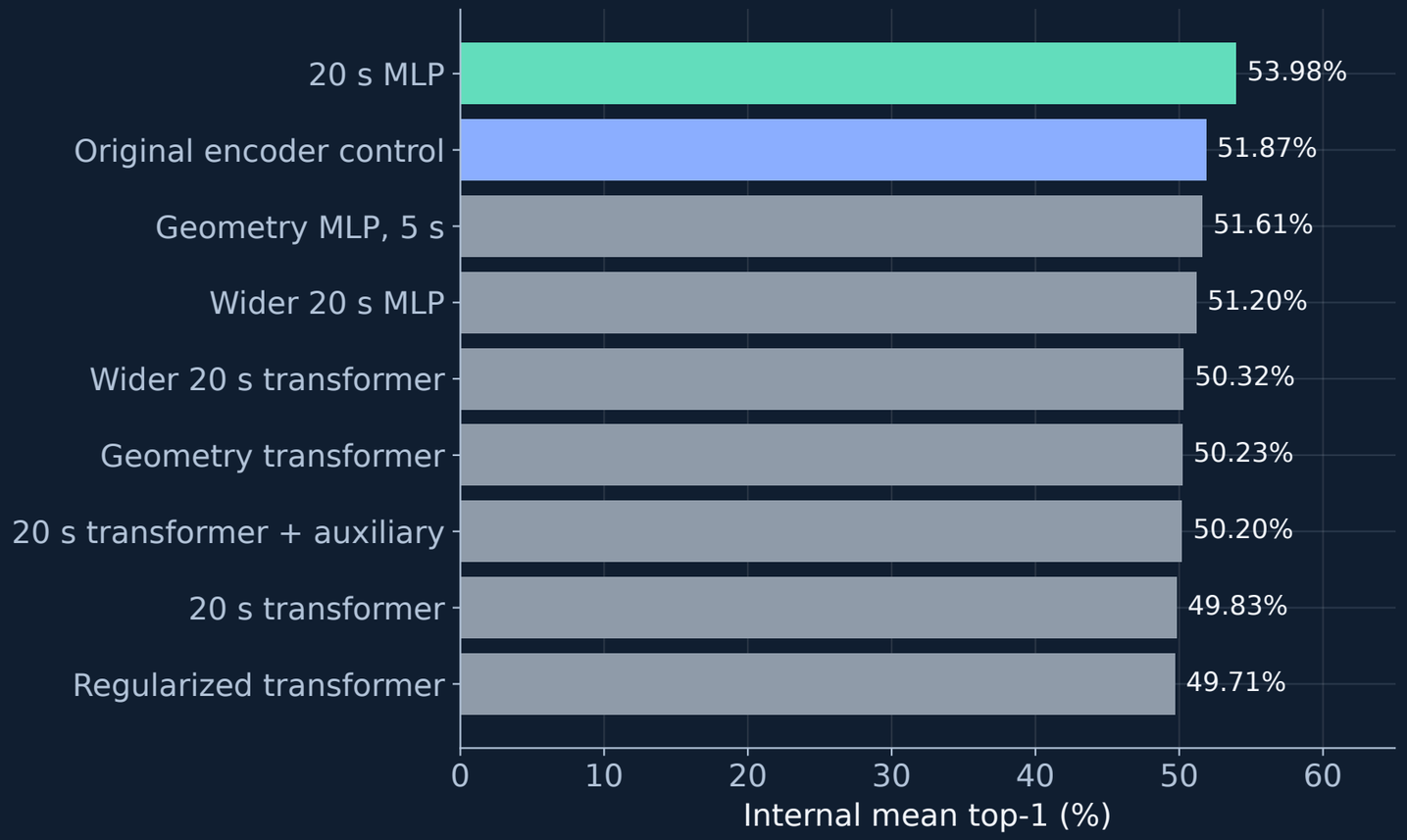
Improved ensemble
Mean confidence 60.83% / accuracy 63.48%

INTERPRETATION

Less overconfidence in aggregate.
Calibration still varies by confidence bin.

The development search also records unsuccessful changes

First internal screen: mean top-1 across horizons. These are not the outer-validation scores.



WHAT THE SCREEN SAYS

More geometry or a wider transformer did not reliably help. The 20 s MLP warranted follow-up.

WHAT IT DOES NOT SAY

An isolated causal contribution for every change. Settings and training procedures differ.

initial_search.csv | Internal split 0 | Development validation; test reserved

The final gain comes with more history and computation

Freeze the selected procedure before evaluating outer validation.

Step	Protocol
Fit / stop / select groups	147 / 21 / 42, inside the original 210 training groups
Confirmation	Two internal partitions; NLL and macro-F1 eligibility guards
Final refit	Four variants × seeds 42, 43, 44; fixed epochs and temperatures
Ensemble weights	Uniform arithmetic probability average across 12 members

ORIGINAL TRANSFORMER → ENSEMBLE

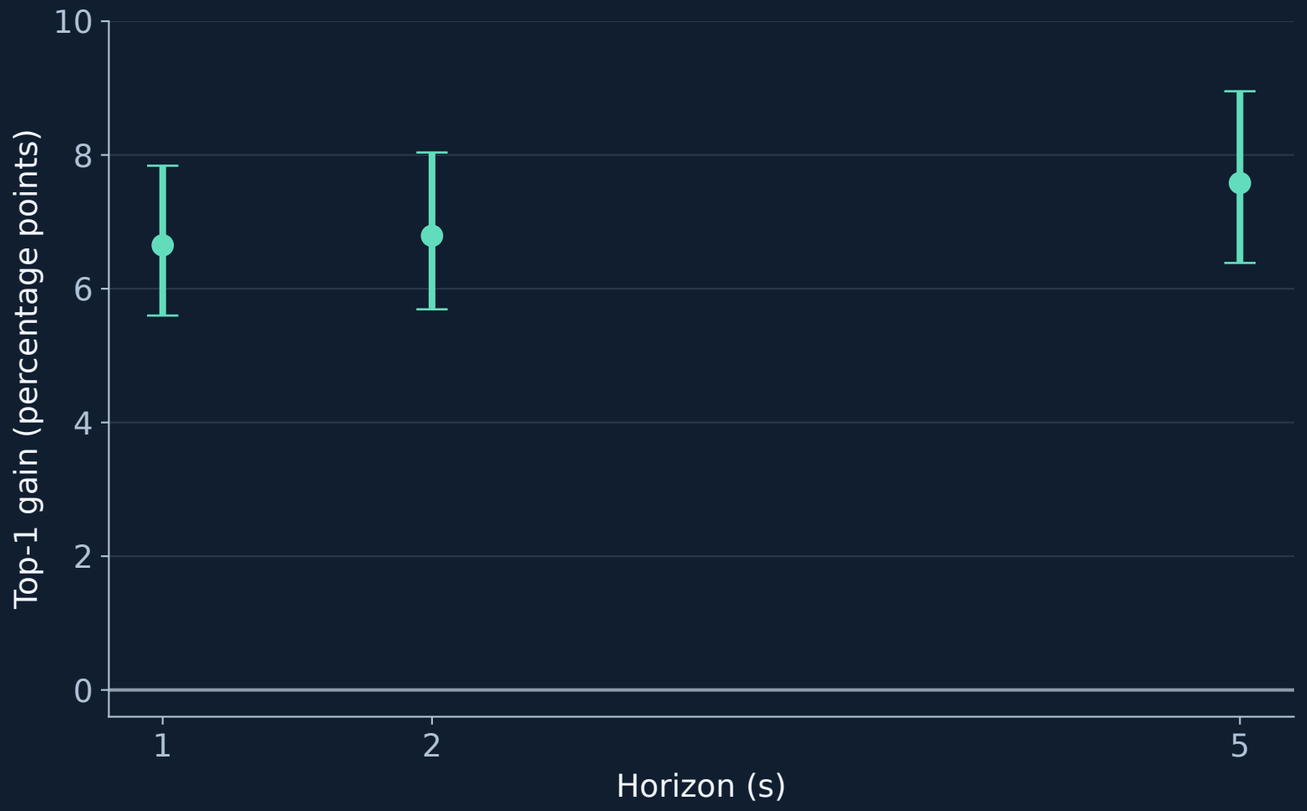
5 s → up to 60 s history
0.44M → 28.98M total parameters

SMALLER MIXTURES REMAIN PLAUSIBLE

Internal accuracy: 60.96% vs 60.98%.
Two vs four variants; one seed each.

The gain remains positive within scenario-level intervals

Resample whole scenarios, preserving all ten pilots and overlapping windows.



PAIRED 95% INTERVALS

+1 s: +6.65 points [5.60, 7.84]
+2 s: +6.79 points [5.69, 8.04]
+5 s: +7.58 points [6.39, 8.95]

THE EVIDENCE LIMIT

45 scenario groups; 5,000 draws.
Original validation already informed baselines.
Reserved test remains unevaluated.

A stronger forecaster, with specific remaining limitations

Use the complete comparison to judge the gain—not only the headline accuracy.

WHAT IMPROVED

Top-1 and top-3 at every horizon.
Catalogue macro-F1 and probability scores.
All ten pilots; changed-selection accuracy.
Net corrections exceed new errors.

WHAT REMAINS UNRESOLVED

Rare and absent labels.
Some unchanged-subset metrics worsen.
Compute cost and minimal ensemble size.
Unseen pilots and realistic sensor inputs.

Next: freeze the comparison, evaluate the reserved test once, and report all metrics.

Complete scorecards, every catalogue entry and reproducible evidence accompany this deck.

How each RL pilot learned: training and validation

Upstream tactical adaptation of the ten frozen corpus pilots | Own episode return; higher is better

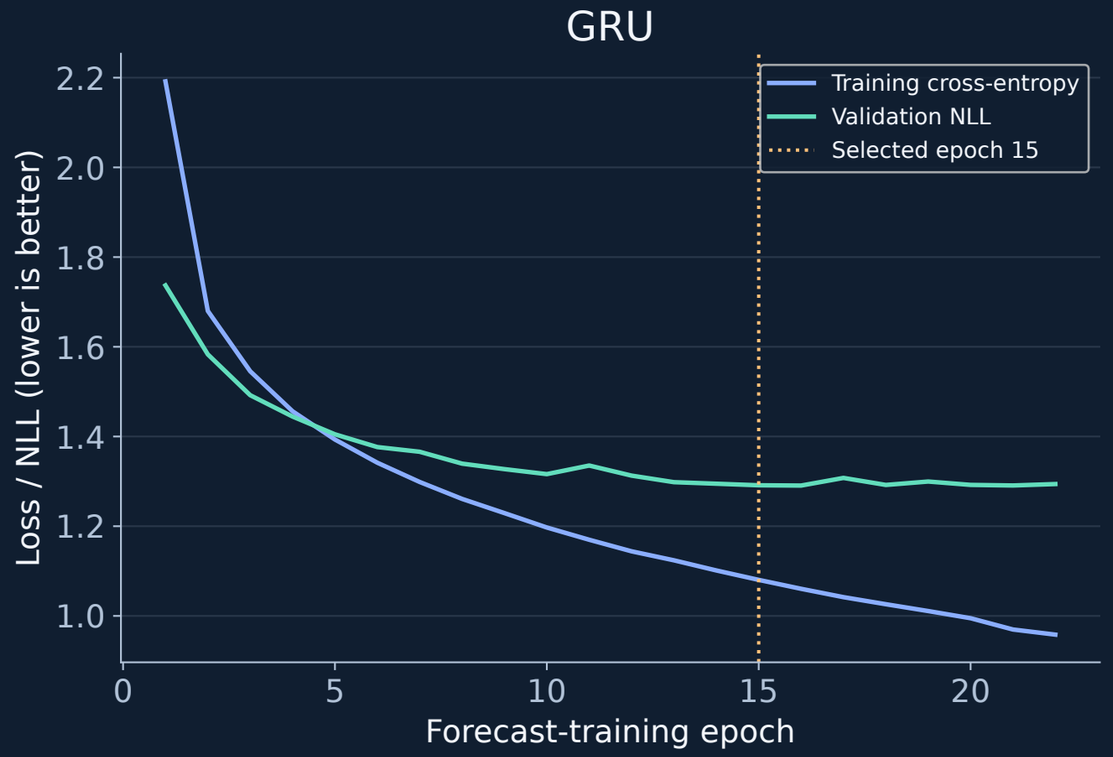
BLUE: training (75 fresh cases / epoch) GREEN: validation (50 fixed cases) STAR: selected checkpoint

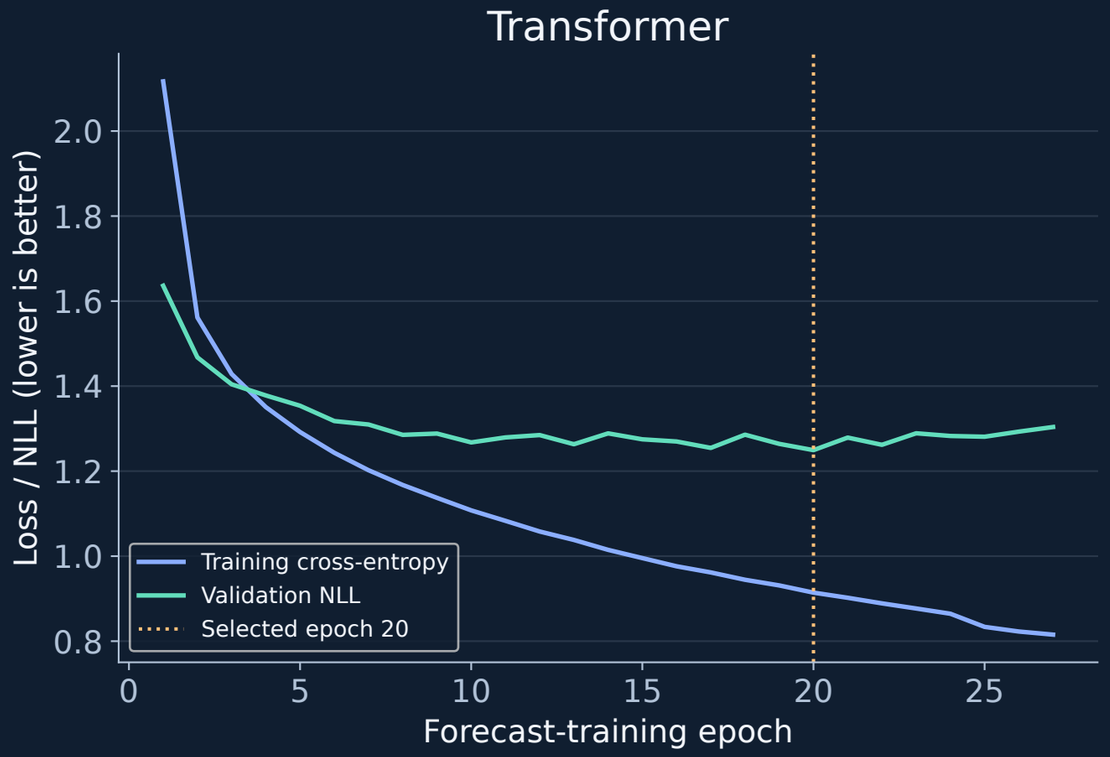


Panels use individual reward scales: compare each pilot with itself. These are flight-learning curves, not forecasting accuracy.

How the original skill forecasters learned

One shared forecaster across all ten pilots | Training loss versus development-validation NLL





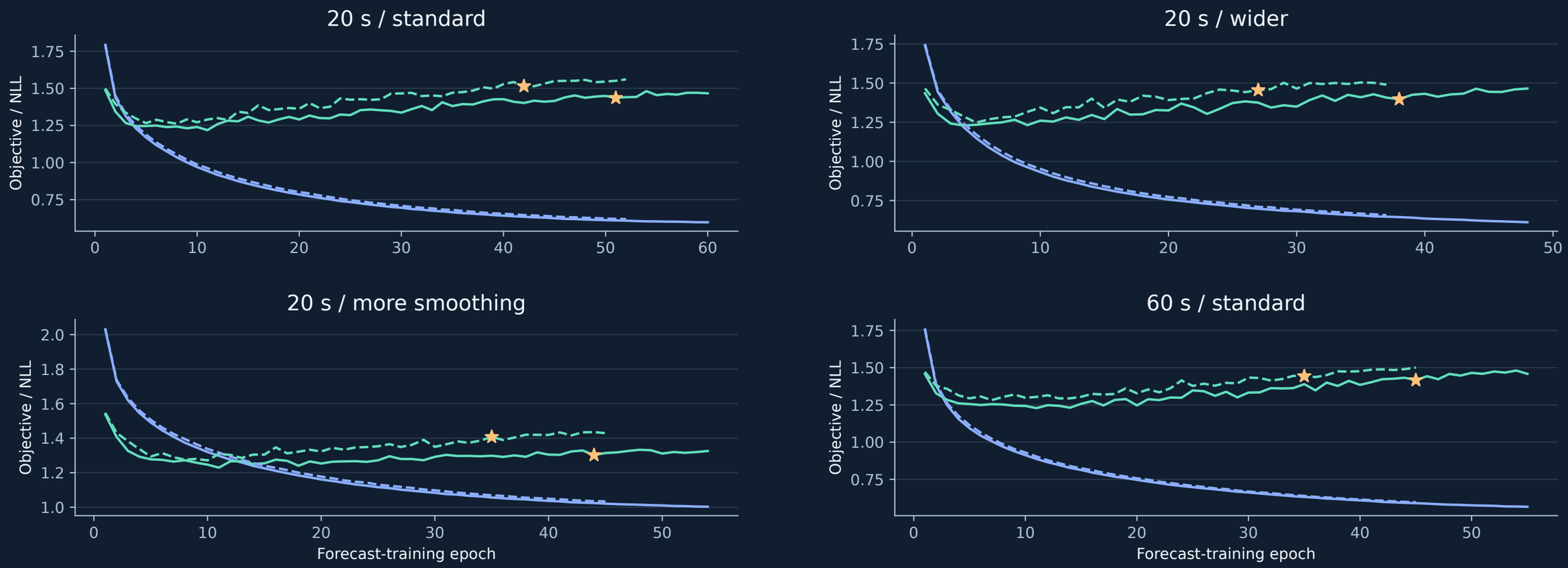
READ THE DIVERGENCE

Training loss continues down; validation stops improving. Keep the selected checkpoint, not the final epoch.

Improved MLPs: learning inside the training split

Four selected variants, both internal partitions | Seed 42 | 147 fit groups and 21 stopping groups

BLUE: training objective GREEN: internal validation NLL SOLID / DASHED: splits 1 / 2 STAR: accuracy-selected epoch

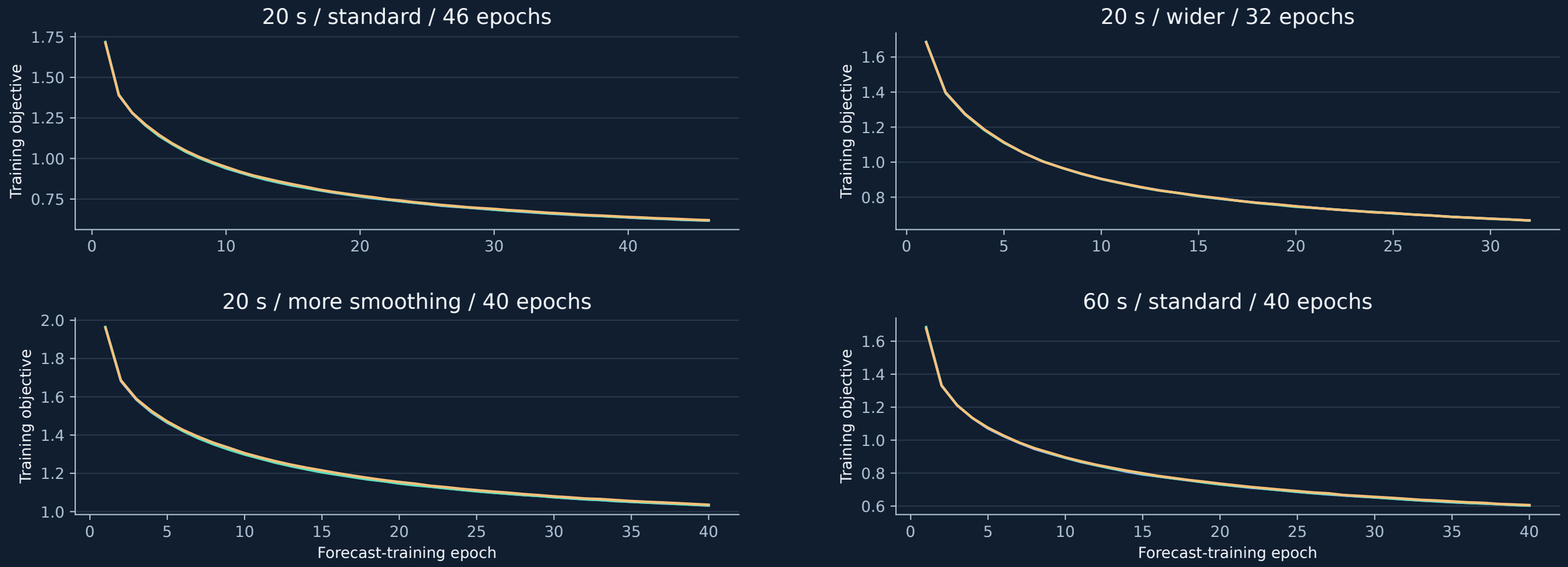


Training uses label smoothing; validation NLL does not. Their vertical gap is not a like-for-like loss comparison.

Final ensemble refits: twelve training curves

Four variants × three seeds | All 210 training groups | Fixed epoch counts

BLUE: seed 42 GREEN: seed 43 GOLD: seed 44



No final-refit validation curve was logged: epoch budgets were fixed in advance. No per-pilot forecast epoch curves were saved.

Temporal MLP: a history window becomes one table row

Temporal = past values in ordered columns | Feed forward = input → hidden layers → output

ONE ORIGIN: THREE TIMESTAMPS

Toy example with two features; values are illustrative.

Time	Speed (m/s)	Range (km)
t − 2 s	250	12
t − 1 s	255	11
t	260	10

FLATTEN → ONE ORDINARY ROW

[250, 12, 255, 11, 260, 10]

Columns: speed(t−2), range(t−2), …, range(t)

Many origins: [N, 3 timestamps, 2 features] → X = [N rows, 6 columns]

Actual 20 s model: [N, 26, 91] + availability flags → [N, 2,392]

Actual 60 s model: [N, 66, 91] + availability flags → [N, 6,072]

Flat row → dense residual layers → 99 scores → 3 horizons × 33 skill probabilities

Inputs stop at t. Targets are labels at t+1, t+2 and t+5. The MLP has no recurrent state.

Two learning problems, two sets of networks

The forecaster learns the behaviour of frozen trained pilots; it does not control the aircraft

1 TRAIN THE PILOTS

PPO actor–critic learns flight decisions.
10 independent reward profiles.
150 × 75 × 10 = 112,500 training flights
per complete population training stage.

2 RECORD FROZEN PILOTS

Select a checkpoint for each pilot.
300 new scenarios × 10 pilots = 3,000 encounters.
B01 observes; A01 selects catalogue skills.
Record observations and selected labels.

3 TRAIN THE FORECASTER

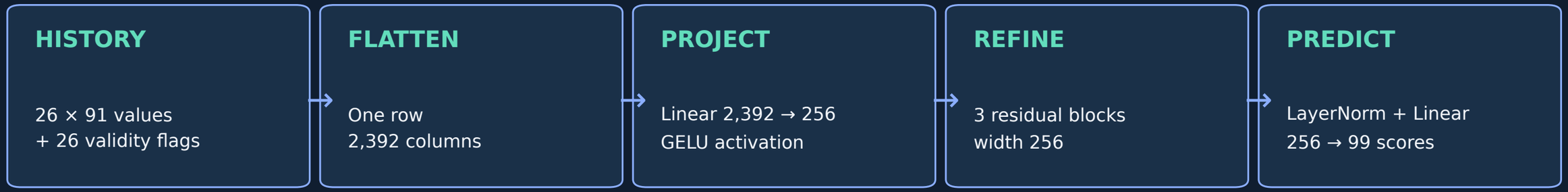
Supervised cross-entropy; no PPO or critic.
One shared model learns across all ten pilots.
Inputs stop at t; labels are at t+1, t+2, t+5.

WHAT THE SCORE MEANS

Predictability of the selected skill label.
It is not an RL reward or combat-success score.
Current evidence: new scenarios, known pilots.

Inside one skill-forecasting network

Worked example: standard 20-second MLP | A supervised predictor of frozen pilots' future selections



Inside each residual block: x + [LayerNorm → Linear 256→512 → GELU → Dropout → Linear 512→256 → Dropout]

99 scores → 3 horizons × 33 skills → temperature scaling → softmax per horizon

WHAT LEARNS?

Weights learn to predict logged future skills.
Cross-entropy averages the three horizons;
unavailable future labels contribute no loss.

WHICH NETWORK IS THIS?

This is the observer's forecasting MLP.
The RL pilot's actor–critic is a separate model.
Pilot weights stay frozen during forecasting.

Four variants, twelve separately trained members

Same 91 features and three residual blocks | Each variant uses seeds 42, 43 and 44

Variant	Input columns	Block width	Params / model	Epochs	Temperature
20 s / standard	26 × 92 = 2,392	256 → 512	1.429 M	46	1.465
20 s / wider	26 × 92 = 2,392	512 → 1024	4.430 M	32	1.401
20 s / smoothing	26 × 92 = 2,392	256 → 512	1.429 M	40	1.169
60 s / standard	66 × 92 = 6,072	256 → 512	2.371 M	40	1.424

Total: 28.98 million parameters across twelve models

CALIBRATE, THEN AVERAGE

Temperature softens each model's confidence.
Average probabilities equally: 1/12 per member.
Each variant contributes 25% of the ensemble.

DIVERSITY HAS A COST

Widths, regularisation and history differ.
9 members use 20 s; 3 use 60 s.
More members are not proven necessary.

Training time: what the saved timers actually measure

Recorded fitting stages on the local GPU setup | Model fitting is separate from simulated flight generation

Stage	Runs	Epochs summed	Timers summed	Per-run range
Original GRU + transformer	2	49	1.50 min	31.1–58.6 s
v2 internal search + confirmation	11	217	5.87 min	17.5–52.9 s
v3 internal search + confirmation	11	466	8.89 min	24.8–71.7 s
Final ensemble refits	12	474	15.42 min	55.3–100.2 s

Recorded fit-stage total: 31.67 min; final twelve refits: 15.42 min

RECORDED ENVIRONMENT

NVIDIA GeForce RTX 4090; CUDA 13.0.
AdamW, batch 1,024, CUDA bfloat16.
Final members take approximately 55–100 s each.

TIMING BOUNDARY

Not the complete experiment duration.
Excludes simulation, preparation and data loading; reporting, audits and selection add further time.

36 completed run summaries; copied validation summaries excluded | Development validation; test reserved

Experiment evolution: why the recipe changed

Top-1 and NLL below are means across +1 / +2 / +5 s | Compare rows only within the same evaluation partition

Experiment	Evaluation	Mean top-1	Mean NLL	Change
Original GRU	Outer validation	53.32%	1.291	5 s / 54 features
Original transformer	Outer validation	57.75%	1.249	5 s / 54 features
v2 control	Internal split 1	51.87%	1.384	Matched internal reference
v2 temporal MLP	Internal split 1	53.98%	1.396	20 s / geometry; NLL stopping
v3 long-history MLP	Internal split 1	58.13%	1.441	Accuracy stopping + calibration
v3 four-variant mixture	Internal split 1	59.09%	1.309	Uniform probability averaging
Final 12-model ensemble	Outer validation	64.76%	1.137	Frozen recipe; three seeds each

Individual accuracy gains came with worse NLL; probability averaging improved both internally.

Two-partition confirmation: four-variant mixture 60.98% vs wide/long pair 60.96% mean top-1.

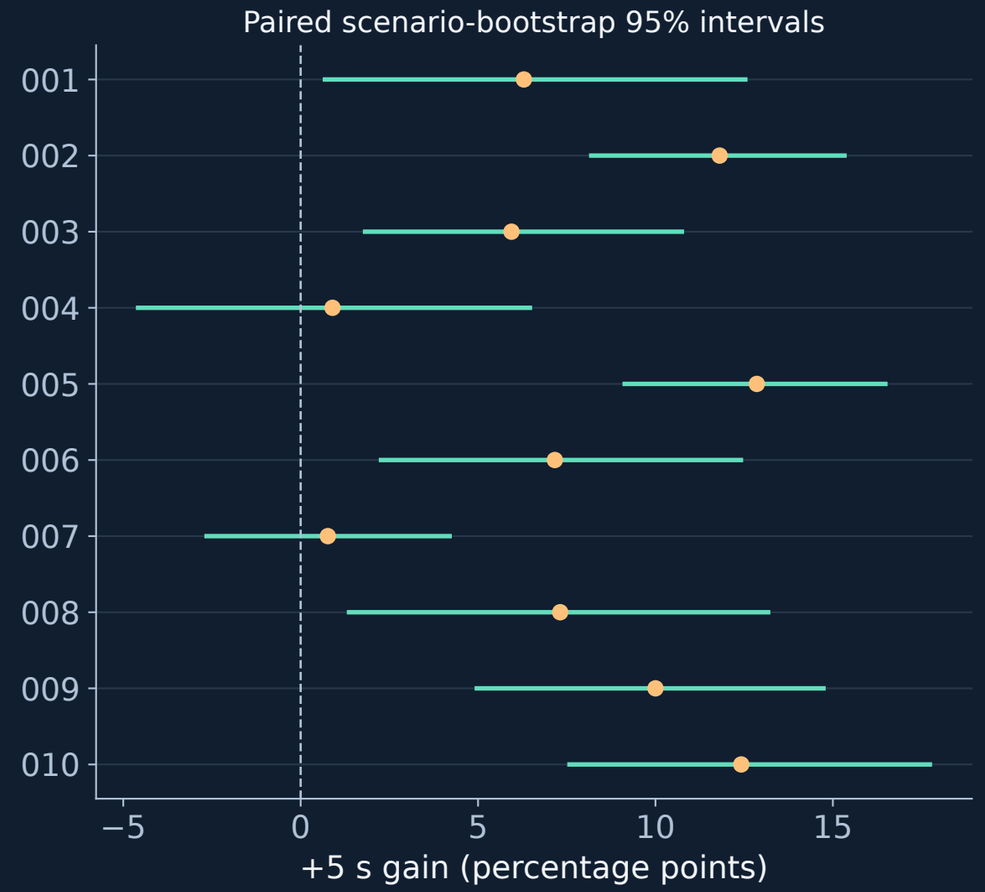
This was adaptive development with several changes together; it is not a clean ablation of architecture alone.

Saved v2/v3 searches, mixture confirmation and fixed validation reports | Development validation; test reserved

Every pilot: forecast accuracy and uncertainty

Ensemble accuracy at all three horizons; paired +5 s gain against the original transformer

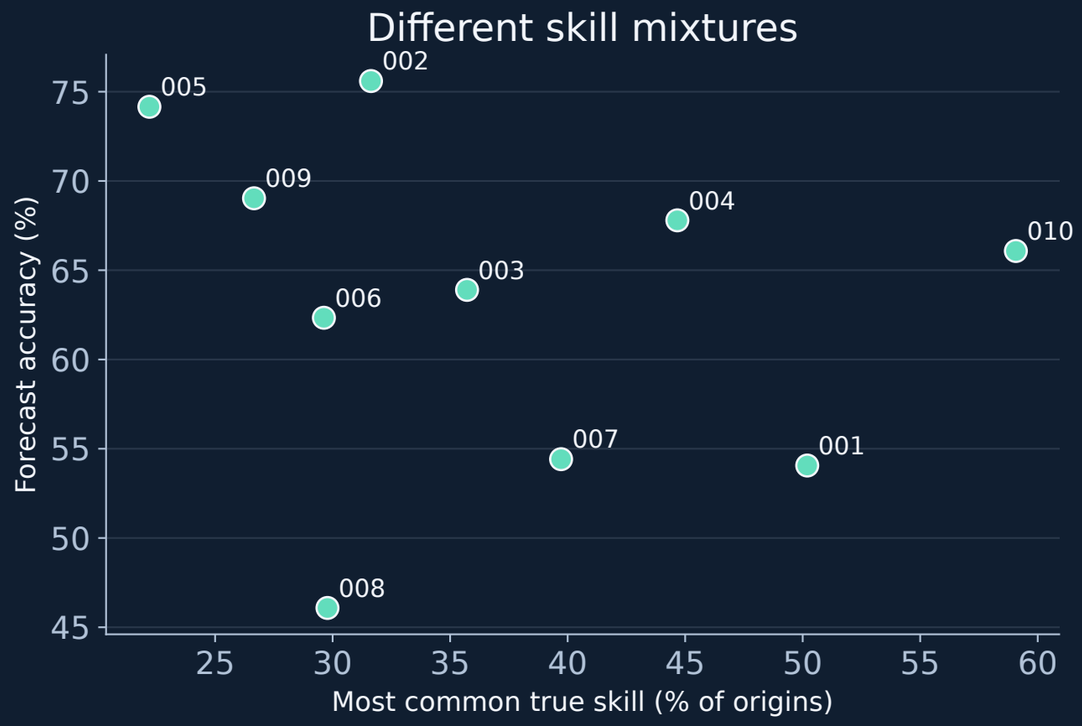
Target pilot	+1 s	+2 s	+5 s
001	54.0%	53.8%	54.1%
002	78.9%	78.5%	75.6%
003	65.6%	65.0%	63.9%
004	68.5%	68.2%	67.8%
005	77.0%	76.8%	74.2%
006	66.2%	65.2%	62.3%
007	57.8%	57.2%	54.4%
008	48.2%	47.9%	46.1%
009	70.3%	70.1%	69.0%
010	68.1%	67.6%	66.1%

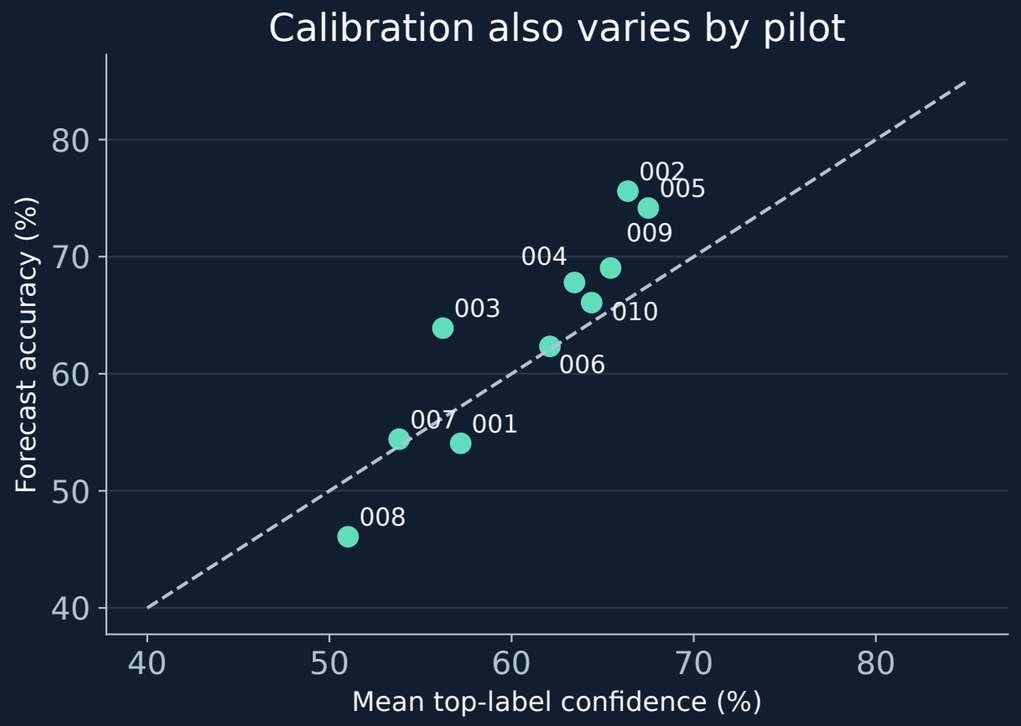


All point estimates improve; uncertainty differs by pilot. Each pilot has 45 validation encounters.

Pilots differ in skill mix and forecast confidence

A high forecasting score is not a rating of pilot competence | +5 s validation



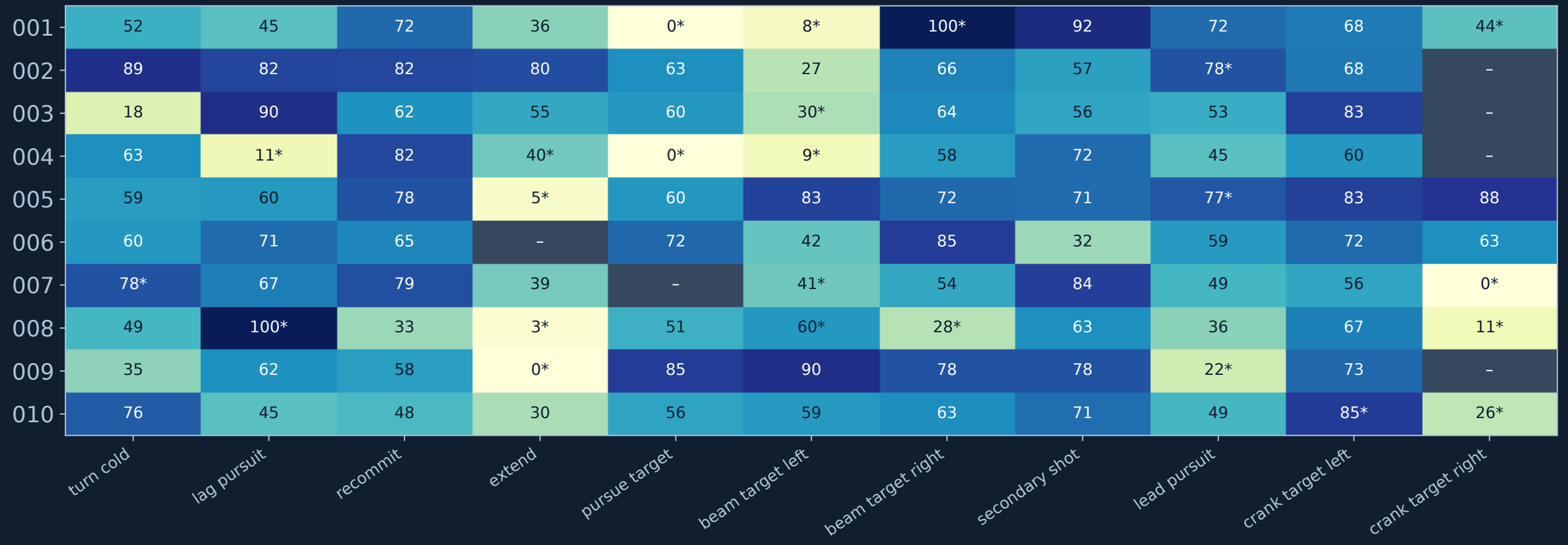


Per-pilot ECE ranges from 2.00 to 10.91 points; pooled ECE alone hides this variation.

Compare label support, change rate and F1 alongside accuracy. These descriptive relationships do not establish causation.

Which skills can we forecast for each pilot?

Recall at +5 s for the eleven most-supported labels | Cell text: percentage; dash: no truth support



* Fewer than 50 true origins in that pilot/skill cell. Full precision, recall, F1 and support are in the companion tables.

Per-skill results need precision, recall and support

The same +5 s ensemble predictions can look different under each metric

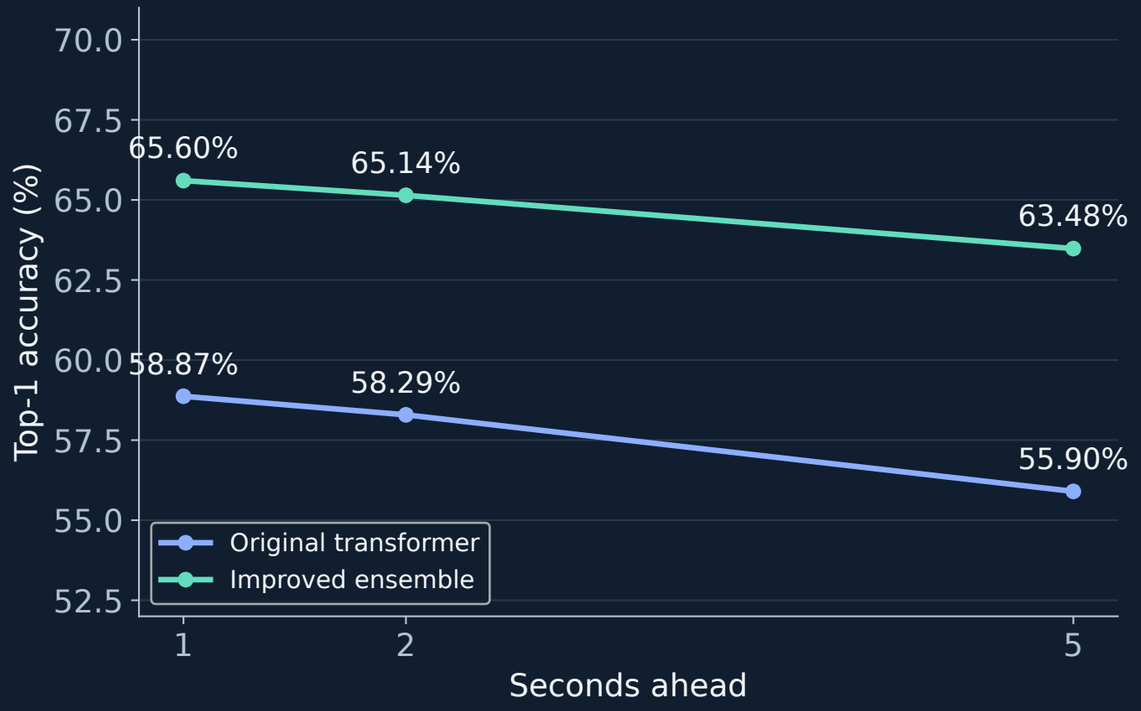
True skill	Origins	Encounters	Precision	Recall	F1
turn cold	5,437	133	0.698	0.660	0.679
lag pursuit	4,872	155	0.605	0.623	0.614
recommit	4,857	168	0.632	0.669	0.650
extend	2,676	89	0.657	0.538	0.592
pursue target	2,561	106	0.539	0.594	0.565
beam target left	1,693	143	0.714	0.710	0.712
beam target right	1,445	138	0.742	0.689	0.714
secondary shot	1,414	153	0.619	0.677	0.647
lead pursuit	1,350	94	0.516	0.564	0.539
crank target left	1,150	183	0.613	0.689	0.648
crank target right	947	56	0.817	0.765	0.790

ASK THREE QUESTIONS

Does it miss this skill? → recall Does it over-predict it? → precision Are there enough independent cases? → support

Forecast horizon: compare the same starting observations

29,458 origins have labels at all three horizons | History length and forecast horizon are different



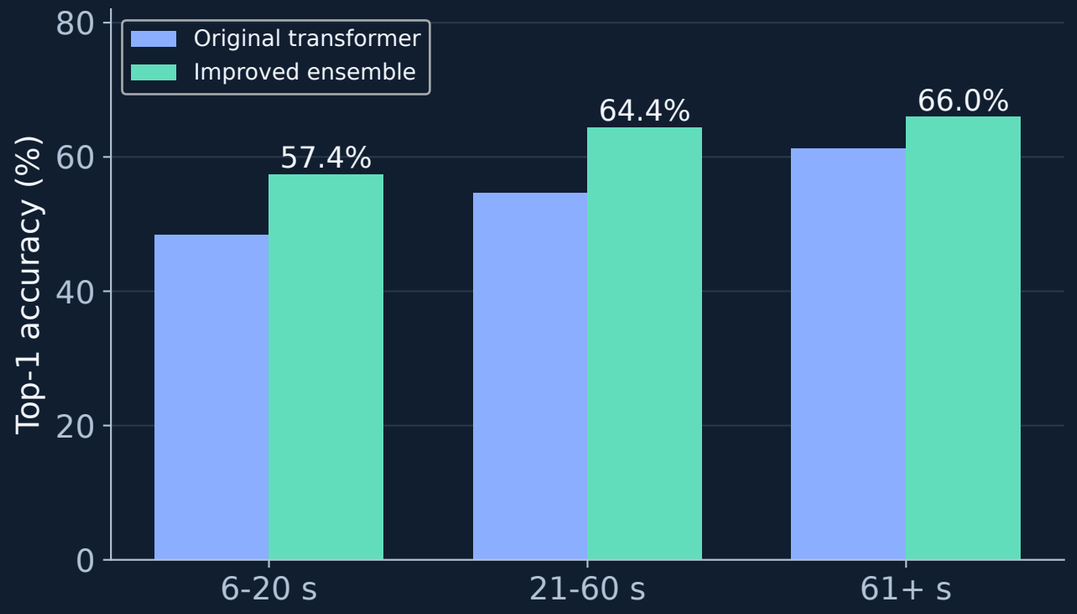
Horizon	Eligible origins	Changed label
+1 s	30,802	7.35%
+2 s	30,466	13.96%
+5 s	29,458	29.39%

Right: each horizon's full eligible set.
Change means endpoint label differs from current.

On matched origins, longer-horizon accuracy falls. Changed-label subsets have different members at each horizon.

Later encounter accuracy is higher—but the cases change

+5 s forecasts, using the common-origin population | Elapsed time is measured at the forecast origin



Origin time	Flights	Groups	Changed
6-20 s	450	45	40.0%
21-60 s	389	39	32.7%
61+ s	225	24	20.2%

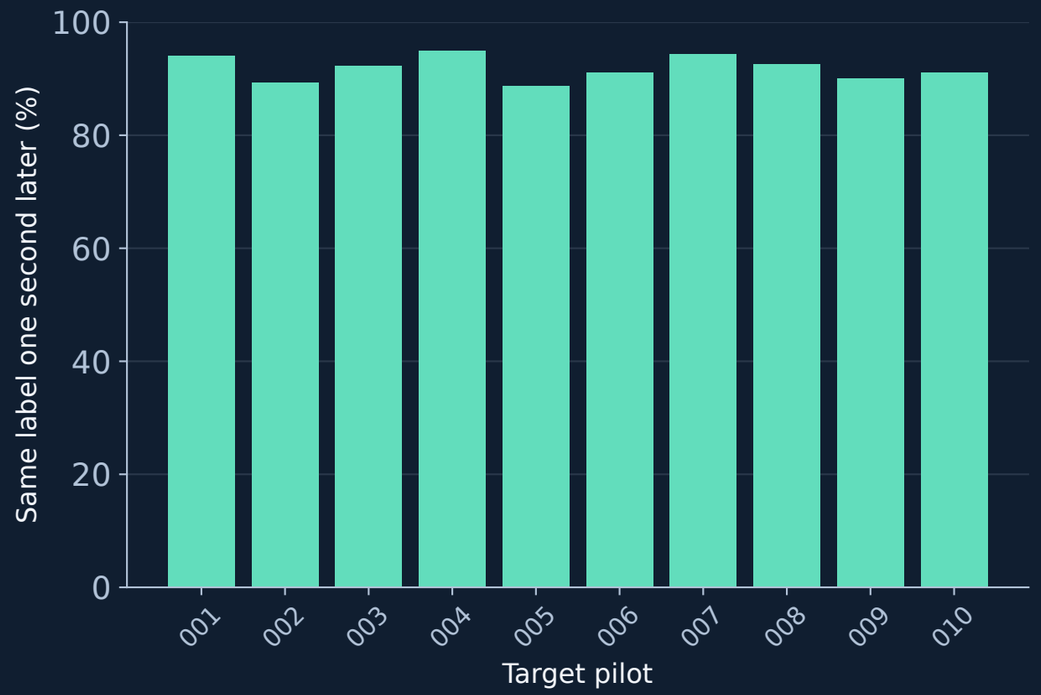
WHAT CHANGES WITH TIME?

Available history, skill mix and surviving encounters. Supported labels fall from 24 to 23 to 17.

Late top-3 slightly worsens: 90.48% → 90.06%. This is not an isolated test of the benefit of longer history.

Skill selections persist, then switch

Full validation decision streams: 38,181 choices at 1 Hz | These are label runs, not physical manoeuvre durations



From	To	Switches
secondary shot	crank target left	140
crank target left	secondary shot	84
beam target left	pursue target	74
crank target left	beam target left	66
pursue target	beam target left	61

Five most frequent observed changes; self-transitions excluded.

91.77% repeat at adjacent decisions | 3,104 switches | 450 episode-ending runs censored

Completed runs only: median 5 s, 90th percentile 16 s. These omit incomplete endings.

Exact catalogue labels can share visible guidance

The task predicts the selected symbolic label; visible motion may not uniquely identify it

Catalogue examples	What the inspected execution code has in common
lag pursuit / extend / turn cold / recommit / pursue target	Can emit the same zero-change guidance fields
beam target left / crank target left	Both emit a −90° heading-change command
launch / secondary shot / short range attack	All set the shoot flag; selected labels remain distinct

INTERPRETATION

A label error can occur even when predicted physical behaviour is similar.
This is an observability question, not proof of a known accuracy ceiling.

Keep exact-label scores. Any behaviour-family score needs a separately justified, frozen mapping.

A reporting structure that makes the claim reviewable

Lead with the task and independent cases; then show the headline and its important breakdowns

Layer	What to report
Scope	Known frozen pilots; new scenario groups; idealised B01 tracks; exact future selections
Headline	Top-1 / top-3, macro-F1 (33 and supported), NLL, Brier, ECE; supports and paired intervals
Pilot breakdown	All 10 pilots × 3 horizons; class mix, support, calibration and gain uncertainty
Skill breakdown	Precision / recall / F1, predicted counts, origin and encounter support; absent labels explicit
Temporal breakdown	Common-origin horizons, endpoint changes, elapsed time, repeated-label runs and censoring
Reproducibility	Split IDs, source hashes, selection chronology, seeds, cost and preserved failed candidates

+5 s accuracy: pooled origins 63.48% | equal encounters 60.85% | equal pilots 63.34%

The chosen weighting changes the question. Keep one primary definition and publish the alternatives.

Next experiments: isolate causes before adding complexity

Proposed work only | Preserve this snapshot and use internal training-group partitions

Order	Question	Controlled test
1	Data-size learning curves	Vary fit groups; fixed stop/select sets; repeat seeds. Test whether more diverse cases still help.
2	Matched feature/history ablations	54 vs 91 features; 5/20/60 s histories. Keep model and training budget controlled.
3	Selection and calibration	NLL vs accuracy stopping, before/after calibration. Report accuracy, F1 and probability scores.
4	Six models versus twelve	Two vs four variants nearly tied internally (one seed each). Test the three-seed refit recipes.
5	Rare skills and transition errors	Broader targeted training cases; cautious reweighting. Preserve natural-prevalence evaluation.

Acceptance: paired accuracy gain, no material F1 / NLL regression, stable pilot breakdowns, and stated compute cost.

Broader claims need separate tests: unseen pilots, realistic sensor errors and genuinely new scenario conditions.